🚴 Cyclistic Bike-Share Case Study Analysis

How Does a Bike-Share Navigate Speedy Success?

1. Executive Summary

This case study analyzes historical ride data for Cyclistic bike-share to identify key differences in usage patterns between **Annual Members** and **Casual Riders**. The goal is to provide data-driven recommendations to convert casual riders into annual members to maximize company profitability.

2. Business Question & Ask Phase

- **Core Question:** How do annual members and casual riders use Cyclistic bikes differently?
- **Business Goal:** Design marketing strategies aimed at converting casual riders into annual members.

3. Data Preparation & Processing Phase

- **Data Sources:** the Divvy 2019 Q1 and Divvy 2020 Q1 datasets from the trip data.
- **Tools Used:** SQL (SQLite / BigQuery) for data cleaning and aggregation, Tableau for data visualization.
- **Cleaning Steps:**
    1. Merged Q1 and Q2 datasets into a unified view.

```
create or replace table capstone-project-504011.cyclistic_data.new_2020 as
select ride_id,
rideable_type
,started_at,
ended_at,member_casual
from capstone-project-504011.cyclistic_data.2020_Q1
union all
select
cast(trip_id as string) as ride_id,
'docked_bike' as rideable_type,
start_time as started_at,
end_time as ended_at,
case
```

```
 when usertype='subscriber' then 'member'
 when usertype='customer' then 'casual'
 else usertype
end as member_casual
from  capstone-project-504011.cyclistic_data.2019_Q1
```

2. Calculated ride_length_minutes from trip start and end timestamps.

```
create or replace table capstone-project-504011.cyclistic_data.cleaned_trips as
select
ride_id,
rideable_type
started_at,
ended_at,
timestamp_diff(ended_at,started_at,SECOND) as ride_in_seconds,
round(timestamp_diff(ended_at,started_at,second)/60,2) as ride_in_minuts
,format_date('%a',date(started_at)) as day_of_week
,member_casual
from capstone-project-504011.cyclistic_data.new_2020
```

3. Extracted day_of_week to examine weekly trends.

```
format_date('%a',date(started_at)) as day_of_week
```

4. Removed records with invalid durations (\le 0 minutes or < 1 minute).

```
create or replace table capstone-project-504011.cyclistic_data.final_cleaned_data
as
select
ride_id,
rideable_type,
started_at,ended_at,
ride_in_seconds,ride_in_minuts,
member_casual,day_of_week
from capstone-project-504011.cyclistic_data.cleaned_trips
where ride_in_seconds >=60
and ride_in_seconds <=86400
and ride_id is not nu
```

5. Aggregated totals and averages by user type (member_casual).

```
select
member_casual
count(ride_id) as number_of_rides,
round(avg(ride_in_minuts),2) as avg_ride_length_in_minuts
```

```
from capstone-project-504011.cyclistic_data.final_cleaned_data
group by member_casual
```

4. Key Insights & Findings

1. **Total Rides Volume:**
    - **Members** account for the vast majority of total rides (~716K rides vs ~67K for casuals in this dataset), showing high daily usage.

```
create or replace table capstone-project-
504011.cyclistic_data.analysis_summery_overall
as
select case
 when member_casual in ('Subscriber','member') then 'Member'
  when member_casual in ('Customer','casual')then 'Casual'
end as member_casua,
count(ride_id) as number_of_rides,
round(avg(ride_in_minuts),2) as avg_ride_length_in_minuts,
round(max(ride_in_minuts),2) as max_ride_length_in_minuts ,
round(min(ride_in_minuts),2) as min_ride_length_in_minuts
from capstone-project-504011.cyclistic_data.final_cleaned_data
group by member_casual
```

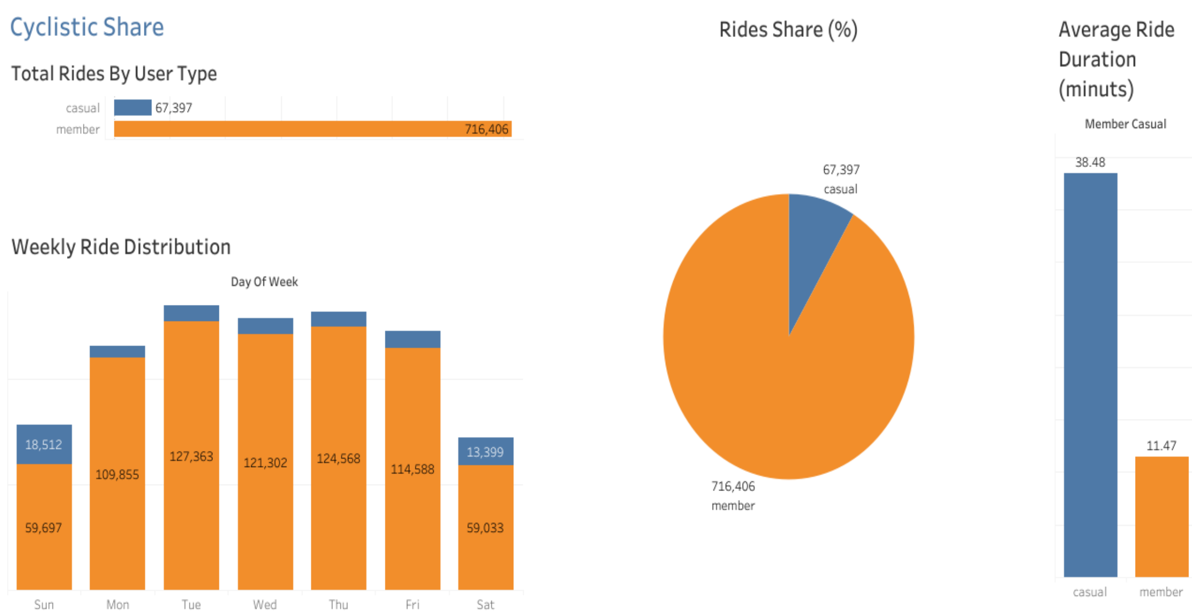
2. **Ride Duration (Time Spent):**
    - **Casual riders** ride for significantly longer durations per trip (average **~38.5 minutes**) compared to **Members** (average **~11.5 minutes**).
3. **Weekly Usage Patterns:**
    - **Members:** Highly active during weekdays (Monday to Friday), indicating regular use for work/school commuting.
    - **Casual Riders:** Activity peaks on weekends (Saturday and Sunday) with longer recreational rides.

```
create or replace table capstone-project-
504011.cyclistic_data.analysis_summery_by_day_of_week
as
select
case
when member_casual in ('Subscriber','member') then 'Member'
when member_casual in ('Customer','casual')then 'Casual'
end as member_casual,
day_of_week,
count(ride_id) as number_of_rides,
round(avg(ride_in_minuts),2) as avg_ride_length_in_minuts,
round(max(ride_in_minuts),2) as max_ride_length_in_minuts ,
round(min(ride_in_minuts),2) as min_ride_length_in_minuts
```

```
from capstone-project-504011.cyclistic_data.final_cleaned_data
group by member_casual,day_of_week
order by member_casual,day_of_week
```

5. **Interactive Dashboard**

- **Tableau Public Dashboard:**
- Visualizations include total rides by user type, ride duration comparisons, pie chart share, and weekly usage patterns.



[Click here to explore the interactive Dashboard on Tableau Public](#)

6. **Recommendations**

1. **Weekend & Leisure Membership Options:** Introduce a "Weekend Pass" or "Seasonal Membership" tailored for casual users who ride mostly on Saturdays and Sundays.
2. **Targeted Summer Marketing:** Deploy in-app promotions and ads at popular tourist/weekend stations during peak hours to highlight the savings of an annual membership for long-duration rides.
3. **Tiered Pricing Strategy:** Adjust pricing so that longer casual rides become more expensive relative to an annual pass, incentivizing frequent long-ride casuals to upgrade to annual status.